

Tied Trit-Planes: Constraining PTQTP to a Uniform Nine-Level Quantizer, with a Persistent Folded Format for Disk-Streamed Mixture-of-Experts Serving

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Abstract

PTQTP decomposes LLM weight matrices into two ternary (trit) planes with two free per-group scales. Tying the scales to a fixed ratio of three collapses the decomposition into a single *uniform nine-level quantizer*, a known balanced-ternary identity. To our knowledge, at the time of writing, this work is the first to impose that identity as a constraint inside PTQTP’s solver. The two trit planes then fold losslessly into one 4-bit code plane that we make the *persistent* serving representation: disk bytes, expert-cache bytes, and kernel input are the same 4.0625-bits/weight blocks, consumed in one integer dot pass. For this conjunction (ratio-3 nine-level code, CPU-SIMD kernels, SSD expert streaming, identical persistent bytes) we likewise found no precedent. We apply the construction to the routed experts of DeepSeek-V4-Flash-0731, a 284B-A13B mixture-of-experts model, quantizing in one shot from the released MXFP4 expert weights and streaming experts from SSD on a 64 GB laptop. Against a 4.5-bit `q4_k` baseline, measured one process per fixture with an expert-lossless anchor arm serving the released bytes as reference control, the tied-ternary model matches the official serving API on 5/5 fixtures at step 0 (`q4_k`: 4/5) and 12/14 captured continuation steps (11/14), scores 86 vs. 84 on a 100-item MMLU subset, decodes 6.7% faster in decode phase, and ships 9% smaller files: no detected fidelity difference at these deliberately small evaluation sizes, with every fixture-level difference between the two arms traced to a single measured near-tie cell. The tied fit nevertheless shows higher weight-reconstruction error and worse perplexity, a measured dissociation between proxy metrics and reference fidelity. A cumulative trunk-ternarization ladder and bitwise-pinned aarch64/x86-64 kernels complete the report. All code, formats, and evaluation artifacts are open source in the `fucina` inference stack.

1 Introduction

Serving state-of-the-art mixture-of-experts (MoE) language models on consumer hardware has become a systems problem with a well-understood shape: the routed experts dominate the parameter count (≈ 145 GiB of `q4_k` expert payload in the 153.3 GiB serving file here, beside ≈ 8 GiB of trunk and embeddings), exceed RAM, and must stream from disk, so decode throughput is governed by

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bytes moved per token and quality is governed by what quantization did to those bytes. A standard llama.cpp byte-side choice is 4-bit K-quantization (q4_k) from the llama.cpp ecosystem [28] (2-bit imatrix variants are also published for this model); a July 2026 community gist [31] asserts that “post-training 2-bit/ternary breaks coding on DSV4 experts” and that “good ternary needs QAT.”¹

This report is an evaluation and systems study inside that setting, with three contributions:

1. The tie, applied and measured. PTQTP [1] writes $W \approx \alpha_1 T_1 + \alpha_2 T_2$ with $T_i \in \{-1, 0, 1\}$ and free per-group scales. We tie the scales at ratio three, $\alpha = (3s, s)$: the composite code $c = 3t_1 + t_2$ then forms a *uniformly spaced* nine-level grid with a single scale. The identity itself is not new: it is claimed for ternary-restricted inference hardware in a patent filed in 2019 and granted in 2024 [2] (with loss-aware training of such $\alpha/3^x$ -scaled ternary replicas in a sibling patent [18]), DBQ built (free-scale, nonuniform) nine-level two-branch ternary quantizers in 2020 [3], concurrent work releases balanced-ternary $d=2$ nine-level PTQ artifacts (model cards labeled uniform alongside a pinned codec computing power-curve, $p=1.5$, level spacing); distribution is dequantized FP16, with a separate kernel demonstrating int4-packed computation on uniform levels [4], and the radix-2 analog is established [5, 6], with recent work [7] proving, for its four-level two-binary-plane family, that the uniform grid is the strict special case of its variable grids, and arguing *against* the constraint (free scales fit better). Our contribution is the application and its measurement: imposing the tie inside PTQTP’s alternating solver and characterizing, at deployment scale, what the constraint costs on reconstruction and perplexity and what it buys on reference fidelity and serving.

2. The fold as the *persistent* representation. Because the tied composite is a single 9-level code, the two trit planes fold into one 4-bit code plane. Packing such composites into 4-bit codes for a kernel is likewise not new [4]; pairing trits into nine-valued 4-bit items also appears, as a virtual intermediate inside a compression pipeline that re-encodes before use, in a patent with 2021 priority, granted 2023 [17]. Serving quantized expert bytes verbatim from a persistent pack is also not unique to us: a concurrent GPU serving system stores kernel-native 2-bit-codebook expert packs on NVMe and pinned RAM and serves them without transcode [22]. What we found no precedent for at the time of writing², and claim narrowly, is this conjunction: the ratio-three nine-level *folded composite* as the persistent representation of a CPU-SIMD, SSD-streamed expert store, with `tq2.0_fx4` (a 520-byte block covering four columns $\times$ 256 elements, 4.0625 bits/weight) keeping the bytes on disk, in the expert-cache slab, and at the integer kernel’s input identical, with no transcode at any tier. That identity is what makes an expert-projection miss one contiguous read and lets a block-granular NVMe tier stripe hot expert prefixes over the primary file: these are the properties the speed results in §4.4 rest on. Multi-plane systems in the literature organize computation over plane decompositions behind LUT kernels [5, 15] or distribute dequantized artifacts [4].

3. Behavioral evaluation with a reference anchor. Quality claims rest on *step-level agreement with the official serving API* (greedy decoding, decoded-text prefix agreement on captured fixtures, not token-ID equality; §3), with a descriptive MMLU comparison alongside (collected in one long-lived server process per arm, without the per-item isolation the fixture protocol uses; §6), and with perplexity reported alongside, because a central empirical finding is that the axes disagree (§4.2). The fixtures and this report’s official-API harness lineage originate in the ds4 project [26]

¹<https://gist.github.com/RockmSockmJesus/30a195ccd9b62e981ec2676a99a57b7e/911283719e3f7e76eb9cae810ee6ec758c793a30>

²Throughout, “at the time of writing” means literature, patent, and code searches completed on this report’s completion date, 9 August 2026; the search services and queries are listed in the repository.

(reference-token comparison methodology more broadly predates it, e.g. 24), whose DeepSeek-V4 work our implementation builds on (§3); we extend the methodology with two elements: an *expert-lossless serving arm* (the released MXFP4 expert bytes repacked without any quantization step) that anchors the comparison (an empirical control: on these fixtures other arms’ deficits are consistent with expert-quantization effects, though this is not a bound on hidden error), and a process-isolation protocol motivated by a measured fixture instability (§3).

The setting gives quantization-quality claims little room to hide: a 284B-A13B production MoE, quantized *post-training in one shot from the released checkpoint* (the experts ship as MXFP4; at the time of writing we found no official higher-precision release of this exact post-trained artifact; the separate Flash-Base checkpoint uses FP8 experts but is a different model), evaluated against the model creator’s own serving behavior. All serving runs in *fucina*³, an open-source inference stack written in Zig, CPU-first with Metal/CUDA GEMM offload, in which the quantizer, the storage formats, the expert store, and the evaluation harness of this report are implemented. Everything below is measured on two machines the author owns: an Apple M1 Max (64 GB, experts streamed from USB SSD with an NVMe cache tier) and an Intel i9-13950HX (128 GB installed RAM, NVMe), with the same files and bitwise-pinned kernels on both.

2 Method

2.1 Tied trit-planes are a uniform nine-level quantizer

PTQTP fits, per weight group, $W \approx \alpha_1 T_1 + \alpha_2 T_2$ by alternating updates of the ternary planes and the two scales. Generic nonzero scale ratios yield up to nine *distinct* composite values; ratios 1 and 2 also give uniformly spaced grids, but with five and seven levels; the composite spans nine distinct, uniformly spaced levels precisely when $|\alpha_1/\alpha_2| = 3$ (or $1/3$, exchanging plane labels; signs absorb into a trit plane). We constrain $\alpha = (3s, s)$ and re-fit: the solver alternates plane updates against the single scale s , which is equivalent to fitting the uniform quantizer

$$\hat{W} = s \cdot c, \quad c = 3t_1 + t_2 \in \{-4, -3, \dots, 3, 4\}.$$

The constrained family is a subset of the free family, so at the respective global optima the tied fit’s weight reconstruction error is worse or equal; our alternating solver is not guaranteed to attain either optimum and scales serialize to f16, so the deployed-family comparison is empirical: the solver’s tied fits show higher error than its free fits on every expert stack of this model (per-expert relative Frobenius error ≈ 0.18 tied; the free fit measured lower on every stack in our solve logs, whose distribution we have not released, a reporting gap noted in §6). Solving all 129 expert-projection stacks (43 MoE layers $\times$ 3 projections, each stack holding 256 experts; 33,024 expert-matrix solves in total) of DeepSeek-V4-Flash-0731 takes ~ 64 minutes on an M1 Max; the solver is deterministic, and a post-conversion verifier re-solves sampled experts and byte-compares.

Quantization source matters: we quantize from the *released* expert weights, which for this model are MXFP4 [32] (fp4-e2m1 codes with one e8m0 scale per 32 elements), exactly dequantizable since every value is a small dyadic rational times a power of two. Quantizing from an already-quantized community file (e.g. q4_k) would add a second quantization step and is avoided.

2.2 The persistent folded format

The tied code $c \in \{-4, \dots, 4\}$ needs 4 bits. We store c directly: `tq2_0.fx4` packs four columns’ 256-element blocks into a 520-byte structure (512 code bytes at two codes per byte + four f16

³<https://github.com/matteo-grella/fucina>

column scales), i.e. 4.0625 bits/weight, comparable to `q4_k`’s 4.5 while representing the *complete* two-plane ternary model. Serving is one dot pass: codes decode to $\{-4, \dots, 4\}$ by arithmetic (no tables required), dot against block-quantized activations via integer SIMD (`sdot` on NEON; `vpshufb/vpsignb/vpdpbusd` on AVX2/AVX-VNNI), one float multiply-add per block. The two ISA arms produce bitwise-identical results by construction (integer lane sums are exact and lane-shaped identically; the float schedule is shared) and are pinned by tests.

Three properties follow from “the fold is the file”: (i) an expert-projection miss is one contiguous read; (ii) the in-RAM slab is byte-identical to the file region, so a block-granular second-tier cache (we stripe hot expert prefixes onto NVMe) can serve partial reads without any re-encoding; (iii) resident and streamed serving use the same bytes and kernels, so outputs are bitwise independent of where an expert happened to be cached. The serving stack (a three-tier expert store: pinned RAM / LRU RAM slots / NVMe stripe over the primary file) is shared with every other quantization format in our engine; the format-specific surface is one kernel and one geometry descriptor.

2.3 The expert-lossless reference arm

Because the released experts are MXFP4, a serving arm with *zero expert quantization* exists: a pure byte permutation repacks the released codes and scales into 17-byte/32-element blocks served by the same store (integer kernel via the doubled-e2m1 trick: fp4 magnitudes $\times 2$ are integers $\{0, 1, 2, 3, 4, 6, 8, 12\}$, and the e8m0 block scale absorbs the $/2$ exactly). This arm serves the original expert bytes bit-for-bit. Its role in the evaluation is an anchor: it removes expert-weight quantization while keeping the same engine and Q8_0 trunk, so deficits beyond it in other arms are consistent with expert-quantization effects on these fixtures: an empirical control, not a bound on hidden error. (The trunk, i.e. attention, shared expert, and dense layers, $\approx 5\%$ of bytes, is Q8_0 in all arms, derived from the released fp8; the anchor arm exposes no deficit from it on these fixtures, though without an unquantized-trunk control this neither measures nor bounds trunk error, §4.1.)

3 Evaluation methodology

Provenance. Our DeepSeek-V4 implementation lineage, the source `q4_k` GGUF conversion, and the behavioral fixtures originate in the ds4 project [26]. The fixtures are ds4’s official-API captures (schema `ds4-official-logprobs-v1`; `deepseek-v4-flash`, checkpoint 0731, greedy, thinking disabled, captured 2026-08-03), used unchanged; ds4 also maintains an official-continuation quality harness whose first-token and target-NLL metrics ours parallels. Our additions on top of that methodology are the lossless anchor arm (§2.3), the process-isolation protocol below, and cross-arm reporting of matched continuation depth. `fucina` is an independent Zig engine whose hot matmul/GEMV kernels are its own; its block-format encoders and numeric codebooks are documented operation-for-operation ports from `ggml/llama.cpp`, its DeepSeek-V4 reference numerics and pre-tokenizer are ports from ds4, and its expert-store design follows earlier expert-streaming systems; the repository’s third-party notices itemize this provenance precisely.

Behavioral fixtures. Five prompts (two long-context, 3,353 and 3,844 prompt tokens; three short) with captured greedy continuations. A fixture *passes* if our step-0 token matches the reference; we additionally report matched continuation depth: our engine decodes *autoregressively* (feeding its own tokens), and the score is the number of leading captured steps whose concatenated text is a prefix of our continuation (longest-common-prefix depth, not per-position teacher forcing), so recovery after a divergence does not count (14 steps across the set). Both columns appear in every table; where they disagree, neither is silently preferred; per-fixture detail is in Appendix A.

Process isolation. During the campaign we found that one fixture sits on a decision knife-edge: its step-0 argmax flips with within-process history (fixtures run earlier in the same process) and with ulp-level binary changes, deterministically reproducible in either state. Every cell of the five matrix arms (`mxfp4`, `q4_k`, `tq2_0.fx4`, `R1`, `R2`) is therefore measured under a one-process-per-fixture protocol on the pinned build; under this protocol the baseline’s knife-fixture failure *reproduces* in a fresh process, so the step-0 comparison does not rest on process history. The cross-protocol repeats that motivated this also surfaced a second condition-sensitive cell (one arm’s long-fixture depth; see Appendix A), reinforcing the protocol choice. Single-fixture margins never carry a claim alone. We recommend the protocol for this harness and similarly stateful runtimes: fixture suites run in one process can silently borrow state.

Language-model loss and task accuracy. WikiText-2 [29] teacher-forced evaluation over the first 512 and 2,048 supervised transitions of the plain-encoded test file (no windowing or resets; these are nested supervised-prefix lengths, not independent context depths; both reported for every cross-family ordering claim). Throughout, *ppl* denotes exp of mean per-token negative log-likelihood; when we compare small differences we state the NLL-space figure. Task accuracy: a 100-question 0-shot MMLU subset [33] (validation files, one globally shuffled sample with fixed seed 20260806, not subject-stratified; the released scorer extracts the first A–D character anywhere in the reply, a rule whose one audited misparse does not change any reported total) served through the engine’s OpenAI-compatible endpoint; with $n=100$ the marginal standard errors are ≈ 3.5 points; paired inference uses the item-level discordance table reported with the result.

Speed. Single-stream greedy decode, 32 generated tokens from a chat prompt, decode-phase rate (prefill excluded), interleaved A/B/B/A rounds with warm caches, both arms on same-age freshly built cache tiers where rebuilt and fresh chat-only usage histograms (the released driver logs both arms’ histogram resets; the `q4_k` tier was freshly built in the accepted battery, the `tq2_0.fx4` tier earlier the same day; a same-day fresh-rebuild A/B moved nothing). The headline battery is accepted under a quiet-substrate gate: the first attempt whose per-arm three-round spread is $\leq 5\%$ (rejected attempts retry after 45 minutes); the rule was fixed before the accepted attempt but not publicly registered. Each battery is six runs in order A/B/B/A/A/B. Two further same-day batteries without the gate are released alongside and agree within a point; outputs of runs of the same file are md5-verified identical (a repeatability check within a format, not a cross-format quality claim). We report the min–max band of the rounds. On the x86 laptop we additionally interleave across *configurations* because sustained load thermally throttles the part by up to $\sim 20\%$: all comparative pairs there are adjacent in time.

4 Results

4.1 Quality: behavior first

Table 1 is the paper’s core. Reading it by row:

The anchor is clean on these evaluations. The released-bytes arm reproduces the official API on every fixture at every step (14/14) and posts the best perplexity of any arm at both prefix lengths. On the reported fixture set this measures the joint visible contribution of the engine, the activation quantization, and the Q8_0 trunk conversion, and lets us read other rows’ deficits as consistent with expert-quantization effects: an empirical control on a small fixture set, not a bound on hidden error nor a proof that those components contribute zero error in general.

Experts format	bits/w	Fixtures (steps)	MMLU-100	ppl@512 / @2048
mxfp4 (released bytes)	4.25	5/5 (14/14)	—	4.39 / 3.15
q4_k (imatrix)	4.50	4/5 (11/14)	84	4.50 / 3.20
tied ternary tq2.0_fx4	4.06	5/5 (12/14)	86	5.08 / 3.72
free-scale K=2	4.125*	2/3 shorts (5/6 steps)	—	4.78 / 3.67

Table 1: Expert-quantization quality on DeepSeek-V4-Flash-0731 (trunk identical Q8_0 in all rows). Fixtures: official-API agreement; **mxfp4**, **q4_k**, and **tq2.0_fx4** cells one process per fixture, the free-scale row from one shorts-only process (§3); the format is “step-0 pass count (matched continuation steps)”. Disclosure: every fixture-level difference between **tq2.0_fx4** and **q4_k** (step-0 5/5 vs. 4/5, depth 12/14 vs. 11/14) is the single measured knife-edge cell (§3); excluding it the two arms are identical (4/4 step-0, 11/13 depth); the free-scale row’s fixture evaluation covers the three short fixtures only. R1 (Table 2) also scores 86 on MMLU. * the free-scale pair does not fold; its serving format stores two separate trit planes at 4.125 bits/weight.

Tied ternary shows no detected fidelity difference from q4_k at 4.06 bits. The full picture is deliberately two-sided: tied passes step-0 on 5/5 fixtures to **q4_k**’s 4/5 and matches 12/14 continuation steps to **q4_k**’s 11/14, but both margins are exactly the single knife-edge cell. Excluding that fixture makes this concrete: the two arms are identical, 4/4 at step 0 and 11/13 on matched depth, which is why we claim no detected difference, not superiority, on reference fidelity, at 10% fewer stored bits and the speed advantage of §4.4. On MMLU the scores are 86 vs. 84; the paired table is 82 items both correct, 4 correct only under tied, 2 only under **q4_k**, 12 both wrong (McNemar exact $p = 0.6875$), a result that fails to reject equality without establishing it; no equivalence margin was prespecified. Against the circulating claim that post-training ternarization of these experts “breaks coding”: the code-completion and code-audit fixtures both pass under the tied quantizer on this fixture set.

Perplexity ranks the rows differently. **q4_k** beats tied ternary by 0.5–0.6 ppl at both prefix lengths even though its fixture and MMLU counts are numerically lower (4/5 vs. 5/5, 11/14 vs. 12/14, 84 vs. 86; small differences that establish superiority for neither arm). The ranking pattern recurs in the comparisons of §4.2, which share solver and evaluation infrastructure and are not independent replications.

4.2 The inversion: free scales improve perplexity; behavioral evidence is inconclusive

The free-scale fit is the natural ablation for the tie. We ran it at three scales:

- **Dense 0.6B** (Qwen3-0.6B, f16): free wins mean teacher-forced NLL on the same WikiText-2 text (30.55 vs. 31.68), as the better-fitting family should. (The dense grid uses the dense harness’s NLL protocol; absolute values are not comparable to the 284B rows.)
- **Dense 1.7B** (Qwen3-1.7B, bf16): tied wins NLL by 25% (35.01 vs. 46.62): the free fit’s two independently-rounded f16 scales per group interact badly with this model; the inversion appears.
- **284B MoE experts (the deployment target)**: free wins perplexity at *both* prefix lengths (4.78 vs. 5.08; 3.67 vs. 3.72), yet fails the code-completion fixture the tied model passes at step 0. That fixture is the measured near-tie (§3); on the stable short fixtures the two variants are indistinguishable (free’s evaluation covers the three shorts only).

Model	Fixtures (steps)	ppl@512	ppl@2048	MMLU-100
experts-only (tq2_0.fx4)	5/5 (12/14)	5.08	3.72	86
R1 + attn q/kv	5/5 (14/14)	5.32	4.13	86
R2 + shared expert	4/5 (11/14)	5.68	4.12	—
R3 full trunk	3/5 (9/14)	6.07	4.38	—

Table 2: Ternarization dose-response (cumulative rungs). R1’s larger matched-depth count over the experts-only model (14/14, matching the lossless anchor) is a single-fixture difference (descriptive, not an improvement claim) while halving attention-projection read bytes. R2’s dropped fixture is the knife-edge short (§3); R3 additionally fails the long-context recall fixture (0/4) while the 3,844-token code-audit fixture still passes 4/4. Perplexity worsens overall but is nonmonotone at 2,048 tokens (R2 sits 0.01 below R1) and identifies no boundary. R3 was measured under the earlier protocol (Appendix A). Because each rung adds several tensor classes at once, the ladder localizes sensitivity to the final cumulative bundle, not to a single component.

At the deployment scale the summary is an asymmetric trade rather than a clean disagreement: the free fit is better on the weight-error objective and on WikiText perplexity at both prefix lengths, while on the fidelity axis the two are indistinguishable except on the single near-tie fixture (which tied passes and free fails), evidence we flag as unstable and do not build on. The robust statement: the constraint’s measured perplexity cost has no detectable counterpart on the fidelity axes we measured; the free arm’s task-level behavior is untested, so full behavioral equivalence between tied and free remains open. Still, the direction is noteworthy against the literature. Proxy-objective gaps are documented: Kim et al. [13] report that weighting layer-output reconstruction objectives by end-loss gradients improves quantization on the settings they test, and the direction varies by setting; QuIP# reports a lower-MSE K-means codebook losing end-to-end perplexity to its E8P codebook [14]; two examples do not establish a field-wide direction. Concurrent 2026 work treats the axes as formally distinct: statistically-lossless quantization separates task-level from distribution-level losslessness and formalizes token agreement [19], and a metric study reports that KLD/perplexity proxies lose ranking power within cohort-specific near-baseline regimes [20]; whether our arms occupy an analogous regime is untested. Our data adds a constrained-fit instance on the fidelity axis: a deliberately higher-weight-error fit with no detected fidelity cost on the cells evaluated (tied-vs-free task behavior was not measured) while behind on perplexity. Perplexity integrates calibrated uncertainty over every position; the fixtures measure argmax agreement with the reference where decisions bind. Neither is sufficient alone; we report reconstruction error, perplexity, reference fidelity, and task accuracy as separate axes, joining prior evaluation work that documents accuracy-matched models diverging behaviorally [23] and token-divergence metrics outperforming perplexity [24], and the 2026 formalizations above [19, 20].

4.3 A cumulative trunk-ternarization dose-response ladder

Holding the tied-ternary experts fixed, we extend ternarization inward in three cumulative rungs: **R1** adds the attention read-side projections (q/kv); **R2** adds the shared expert; **R3** ternarizes the full trunk (router, compressors, indexer, residual writers). Table 2.

Two observations. First, *attention read-side ternarization does not reduce fixture agreement*: R1 matches the lossless anchor’s full-step sweep, at +0.24 ppl and with materially smaller attention reads. Second, the established “MoE experts tolerate aggressive quantization” result [10, 11], with component-wise sensitivity studies now including QuantMoE-Bench’s sweep over attention, shared experts, and routed experts [12], is consistent with this five-fixture case of a nine-level quantizer on an LLM-scale MoE (no general extension across MoEs is established): fixture failures beyond

Machine	Config	q4_k	tq2.0_fx4	mxfp4
M1 Max 64GB	gated A/B/B/A, fresh tiers+histograms	2.91–2.95	3.11–3.12	—
M1 Max 64GB	cross-session band	2.46–2.95	2.92–3.24	2.48–2.54
i9-13950HX	evicted cache, NVMe	—	2.62	2.40–2.47
i9-13950HX	page-cache warm	—	3.31–3.33	—

Table 3: Decode throughput (tok/s; 32-token greedy, decode phase). The headline claim is the gate-accepted interleaved battery at the pinned code revision (fresh chat-pure histograms, same-day tiers, per-arm spreads 0.3%/1.2%): **tq2.0_fx4** decodes **+6.7%** over **q4_k** by median of rounds computed from logged unrounded times, with md5-identical outputs per file and -9.2% file bytes (139.2 vs. 153.3 GiB). Two further same-day batteries without the gate give $+6.6\%$ and $+6.7\%$; one **q4_k** round among them measured 1.95 with degraded expert-path timing in its own log (369 vs. ≈ 262 ms/token expert time); its cause is unresolved, the battery median is insensitive to it, and all eighteen raw round logs are released. The cross-session band shows absolute rates are substrate-state sensitive (tier build conditions, free-space headroom, kernel revisions): an earlier-session pair on an older **q4_k** tier gave 2.46–2.51 vs. 2.92–3.01 ($+19\%$); we headline the matched-fresh measurement and report both. The **mxfp4** anchor matches **q4_k**-class speed while serving the original bytes. x86 rows use thermally-adjacent interleaving (§3).

the knife-edge short appear only in the final cumulative rung, under the mixed protocol disclosed above. This is compatible with the newest contrary finding located at the time of writing (Wang et al. [9] report MoE models as *more* ternarization-sensitive than dense when nearly the whole model is ternarized) on the reading that sensitivity concentrates in trunk components, which whole-model treatment inherits; our cumulative ladder cannot, however, identify which component in the final bundle is responsible, and the models and quantizers differ, so we offer this as a consistent reading rather than a reconciliation.

4.4 Speed

The measured decomposition on the matched pair: the **tq2.0_fx4** run reads 30.31 GB from disk per 32-token round against **q4_k**’s 33.89 GB (-10.6% , tracking the bits/weight ratio), completes those reads in 10.7–10.9 s vs. 11.8–12.0 s of cumulative blocked-read time after overlap (the engine’s miss-batch counter), and pins $+10.8\%$ more experts in the identical RAM budget (1,344 vs. 1,213); per-projection misses are single contiguous reads by format construction (§2.2). These byte and pin figures are identical in every round of all three batteries and the earlier-session pair: the format’s mechanical advantage is substrate-invariant even where absolute rates are not. Two further storage-system observations from the campaign, with their configurations: (i) a *cache-budget cliff*: decode collapses when the expert-cache budget leaves insufficient OS page-cache headroom (M1, 64 GB RAM: knee at a 26–28 GB budget; i9, 125 GB RAM: a 90 GiB budget decodes slower than a 24 GiB one, 1.87 vs. ~ 2.2 tok/s under identical eviction protocol); past the knee, *reducing* the budget is faster; (ii) on the hybrid P/E-core part, the optimal thread shape inverts with cache-hit regime (at 76% hit, 24 threads across P+E beat 16 hyperthreads on 8 P-cores, 2.46 vs. 1.83; at 90% hit the ordering reverses, 2.28 vs. 3.05). We report these as observations under our stated configurations rather than general laws.

4.5 Cross-ISA reproduction

The same files serve on aarch64 and x86-64. Integer kernel arms are pinned bitwise-identical (exact lane-sum equivalence between **sdot** and lane-folded **vdpbusd** formulations; shared float schedule); the full test suite passes on both ISAs; the five-fixture behavioral results reproduce exactly on

x86, including the knife-edge fixture. End-to-end language-model loss drifts between ISAs (anchor arm at 512 tokens: perplexity 4.44 vs. 4.39, i.e. 1.1% relative, an absolute difference of 0.0113 nats/token), accumulated ulp drift from architecture-specific schedules in attention and norms, with no behavioral consequence on any fixture. We note this as calibration for what “bit-exact” can and cannot mean across ISAs: kernel-level yes, transformer-stack level no, behavior in practice yes.

5 Related work

PTQTP [1] introduces the dual trit-plane decomposition this work constrains; adjacent ternary-decomposition work factorizes into expanded-rank ternary matrices with learned real scales [8] or advocates variable grids over binary planes [7]. The ratio-3 ternary identity itself has multiple prior instances: geometric ternary weight-set replication $(\alpha, \alpha/3, \alpha/9)$ on ternary-restricted inference hardware [2]; DBQ’s two-branch ternary quantizer, which realizes the nine-level composite with *free* learned scales and a nonuniform grid [3]; concurrent balanced-ternary $d=2$ nine-level PTQ (uniform-labeled artifacts and a power-curve codec; FP16-distributed checkpoints; separate uniform-grid int4 kernel) [4]; and the radix-2 analog [5, 6]. Relative to these, this report contributes the constraint’s application inside PTQTP’s solver, the measured error/perplexity/fidelity trade-off at MoE deployment scale, and the folded code as the *persistent* serving representation (disk, cache slab, and kernel consume identical bytes). Prior multi-plane systems organize *computation* over plane decompositions behind LUT kernels [5, 15]; they persist preprocessed kernel-specific layouts but do not describe this SSD-expert-store identical-bytes invariant, and prior nine-level artifacts distribute dequantized weights [4]. A concurrent MLX system streams MoE experts from disk and executes base-3-packed single-plane ternary experts directly on Metal [30], the closest packed-ternary serving antecedent we found, differing in code geometry (single-plane $\{-c, 0, c\}$ vs. ratio-3 two-plane nine-level) and platform. Expert-quantization robustness is established qualitatively [10], at trillion-parameter scale with a bespoke compressed format [11], and with component-wise precision benchmarking [12]; our ladder adds a cumulative nine-level dose-response on a modern LLM-scale MoE evaluated on reference fidelity and language-model loss. The deployment setting (this exact model streamed from SSD on 64 GB consumer machines with official-API fidelity testing) was demonstrated by ds4 [26], on which our fixtures and conversion lineage rest. Concurrently, vLLM-MoE serves DeepSeek-V4-class MoEs on consumer Blackwell GPUs from persistent kernel-native 2-bit expert packs tiered over NVMe and pinned RAM, with a public history (June–July 2026) that precedes our repository’s; the two public histories and disjoint implementations (different quantization alphabets, block formats, cache hierarchies, and execution stacks: GPU/vLLM vs. CPU-SIMD/Zig) are consistent with, though cannot prove, independent development, and we make no priority claim for persistent low-bit expert serving in general [22]. Within this setting, our deployment result adds tied-ternary experts with no detected fidelity difference from q4.k on the reported evaluations and a higher observed decode rate under the reported protocol (no equivalence or causal format-only speed claim). Single-plane ternary PTQ at MoE scale is contemporaneous [9, 25, 30]; mixed-precision expert offloading uses lower-precision experts on cache misses [16]. Proxy-objective gaps are documented [13, 14]; §4.2 situates our fidelity-axis instance among them.

6 Limitations

Construction. The tie-to-uniform identity has prior instances (§5); this report’s claims are about its application, measurement, and persistent serving format, not its invention. The quantization source is the released MXFP4 checkpoint: exactly dequantizable, but itself the product of the model

creator’s quantization-aware training; at the time of writing no official higher-precision release of this exact post-trained artifact existed.

Evaluation. The fixture set is small ($n=5$ prompts, 14 captured steps whose positions are dependent within a prompt), non-random (inherited from ds4’s captures), and measured against a hosted API whose behavior can change over time; no formal equivalence margin is prespecified; “no detected difference” means exactly that, a non-rejection at these sizes, supported by the paired MMLU analysis and the fixture-exclusion computation in §4.1. The step-0 pass criterion is one of several defensible rules, so we co-report matched continuation depth (tied 12/14 vs. **q4_k** 11/14; identical 11/13 with the unstable fixture excluded). Cross-protocol repeats surfaced two condition-sensitive fixtures: the near-tie short, whose step-0 outcome flips under within-process history and ulp-level build changes, and one long fixture whose **q4_k** depth measured 4/4 in an earlier full-process run on a pre-merge build but 2/4, deterministically on repeat, under the canonical protocol; root causes in our runtime are unexplained. Every cell of the five matrix arms is measured one process per fixture on the pinned build (under that protocol the baseline’s knife-fixture failure reproduces, so the margin is not a history artifact), and the 5/5-vs-4/5 margin is still flagged wherever it appears. R3’s rows (a ladder rung with wide margins) retain the earlier protocol, disclosed in Appendix A. The MMLU subset is a globally shuffled sample, not stratified, and its released scorer’s first-letter-anywhere extraction rule is a documented defect (one audited misparse, no total changed), and all 100 items ran in one long-lived server process per arm without the per-item isolation the fixture protocol uses, so the MMLU comparison is descriptive, not independent corroboration; the free-scale reconstruction-error distribution is not in the artifact bundle. The free-scale arm’s fixture evaluation covers only the three short fixtures. MMLU uses a custom 100-item subset ($\pm 3.6\text{pt}$ s.e.; differences reported are within noise). Perplexity uses fixed token budgets (512/2,048), not the full test set; the cross-family orderings we rely on hold at both depths, while the R1–R2 ordering inverts by 0.01 at 2,048. API agreement is a fidelity metric: for intentionally different quantizers it does not by itself establish downstream superiority, which is why fidelity, perplexity, and task accuracy are reported as separate axes.

Scope. Single model family at the headline scale; the dense-scale grid uses a different family (Qwen3), so scale and architecture are confounded in the dense-to-MoE comparison. The ladder’s rungs are cumulative bundles and cannot localize sensitivity to single components. The trunk is Q8_0 in every Table 1 arm; the ladder’s rungs ternarize trunk tensor groups from those Q8_0/F16 donor forms (one additional near-lossless conversion hop in their lineage), so full-model ternary serving (R3) inherits a converted trunk and the ladder measures ternarization on top of that lineage. Speed numbers come from one engine on two machines the author controls, as medians and min–max bands over interleaved rounds (eighteen same-day rounds across three batteries, six under the acceptance gate) without formal uncertainty estimates, from a single prompt at 32 decode tokens (no prefill, concurrency, tail-latency, or cold-start characterization); because the formats generate different continuations, their routed-expert workloads differ, and the speed comparison does not causally separate format, kernel, cache, and workload effects. Absolute decode rates proved substrate-state sensitive across sessions (an earlier **q4_k** tier built under low disk headroom on a pre-optimization kernel revision measured 2.46–2.51 where the fresh rebuild measures 2.88–2.92), which is why the headline pair holds tier age, disk state, and code revision matched; the byte-read and pin-count decomposition is identical in every round of every session. The **tq2_0.fx4** enum is a **fucina**-specific GGUF extension not parseable by stock GGUF tooling; the sibling two-plane format remains the interchange form. The thermal-interleaving discipline is described so others can reproduce it.

7 Reproducibility

Engine, converter, format, store, and kernel code is in the open-source **fucina** repository (MIT) at commit `d30ce52`; the report repository separately carries the MMLU runner, a minimal isolation wrapper, and the speed-battery and isolation-matrix driver scripts. In **fucina**: the tied solver and converter (`convert-ds4-fp4`, one-shot from the released checkpoint; a 9 GB trunk-only donor makes conversion self-sufficient on any machine holding the public checkpoint), the `tq2.0_fx4` exporter (`export-gguf --ptqtp-native`), the lossless `mxfp4` repack, the expert store with the NVMe stripe tier, the fixture/NLL harness, and the kernel parity tests that pin the two ISA arms; the `tq2.0_fx4` block layout is specified in `docs/PTQTP.md`. The model is `deepseek-ai/DeepSeek-V4-Flash-0731` at revision `7872f01b` [27], whose card describes an attached speculative-decoding module (the artifact totals ≈ 304 B parameters; the served core is 284B/13B-active per the base card and technical report [21], and we do not serve the attached module); the `q4_k` baseline derives from the ds4 project’s imatrix conversion, and the behavioral fixtures are ds4’s published captures (prompts, official-response JSON, and capture scripts in the ds4 repository [26]; a pinned checkout is fetched to `refs/ds4` by `tools/fetch_refs.sh` in **fucina**). The MMLU subset is derived from the validation files with fixed seed 20260806 (globally shuffled sample; the scorer and its extraction rule are released); per-item results for the paired analysis are produced by the harness. Builds: Zig 0.16.0, `-Doptimize=ReleaseFast`, macOS 14.6 (M1 Max) and Linux 6.x (i9-13950HX); serving flags for each reported configuration appear alongside the numbers. This report’s source is maintained at <https://github.com/matteo-grella/tied-trit-planes>.

Acknowledgments

The DeepSeek-V4 serving lineage this work builds on (the model conversion, the SSD-streaming deployment setting, and the official-API fixture methodology) originates in Salvatore Sanfilippo’s ds4 (DwarfStar) project [26]. **fucina** also owes its formats to the `ggml/llama.cpp` ecosystem.

A Per-fixture results

Step-0 pass and matched continuation depth per fixture and arm. Every cell of `mxfp4`, `q4_k`, `tq2.0_fx4`, R1, and R2 is measured one-process-per-fixture on the pinned build (§3); raw harness logs ship in the repository artifacts. R3’s rows (a ladder rung with wide margins) are from earlier full-process and shorts-only runs of the same code. Cross-protocol repeats identified two condition-sensitive cells, both disclosed in the text: `short_code_completion` (the near-tie; step-0 flips with process history and ulp-level build changes; R2’s full-process run passed it, the canonical value is its reproducible isolated FAIL) and `long_memory_archive` for `q4_k` (4/4 in an earlier full-process run on a pre-merge build; 2/4 deterministically under the canonical protocol). The free-scale arm was evaluated in one shorts-only process, its `short_code_completion` executing first (history-free position).

Fixture (prompt tokens)	<code>mxfp4</code>	<code>q4_k</code>	<code>tq2.0.fx4</code>	R1	R2	R3	free
<code>long_code_audit</code> (3,844)	4/4	4/4	4/4	4/4	4/4	4/4	—
<code>long_memory_archive</code> (3,353)	4/4	2/4	2/4	4/4	2/4	0/4	—
<code>short_code_completion</code> (27)	1/1	0/1	1/1	1/1	0/1	0/1	0/1
<code>short_italian_fact</code> (21)	4/4	4/4	4/4	4/4	4/4	4/4	4/4
<code>short_reasoning_plain</code> (18)	1/1	1/1	1/1	1/1	1/1	1/1	1/1
step-0 passes	5/5	4/5	5/5	5/5	4/5	3/5	2/3
matched depth	14/14	11/14	12/14	14/14	11/14	9/14	5/6

Table 4: Per-fixture matched continuation depth (free-scale shorts: 0/1, 4/4, 1/1 = 5/6). A fixture’s step-0 pass is equivalent to depth ≥ 1 on its first step.

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